FINAL PROJECT REPORT

Heart Disease Analysis using Tableau Dashboard

1. INTRODUCTION:

1.1 Project Overview

Heart disease is one of the leading causes of death worldwide. Identifying risk factors and understanding patterns in heart disease data is essential for improving healthcare outcomes. With the increasing availability of healthcare data, data visualization tools can help extract meaningful insights from complex datasets.

This project focuses on analyzing heart disease data using Tableau, a powerful data visualization tool. The dataset contains health-related indicators such as BMI, smoking habits, alcohol consumption, age category, sleep duration, diabetes, and other medical conditions.

By using interactive visualizations, the project aims to identify patterns and correlations between lifestyle factors and heart disease risk. The dashboard provides healthcare professionals and researchers with an easy-to-understand interface to analyze the data and make informed decisions.

1.2 Purpose

The main purpose of this project is to analyze heart disease risk factors and present insights using an interactive Tableau dashboard. The project helps in understanding how various lifestyle habits and health conditions influence the likelihood of heart disease.

The objectives of this project include:

Analyzing heart disease patterns using healthcare data.

Identifying key risk factors such as smoking, BMI, and physical activity.

Creating interactive visualizations using Tableau.

Helping healthcare professionals interpret health data effectively.

Supporting data-driven decision-making for disease prevention.

2. IDEATION PHASE:

2.1 Problem Statement

PS-1

I am a healthcare analyst trying to identify key risk factors contributing to heart disease, but the dataset contains many variables that are difficult to interpret manually. Because the relationships between lifestyle habits and heart disease are not clearly visible in raw data, it becomes challenging to draw meaningful insights, which makes me feel frustrated while analyzing healthcare information.

PS-2

I am a public health researcher trying to understand how lifestyle habits such as smoking, alcohol consumption, sleep time, and physical activity influence heart disease risk. However, traditional data analysis methods do not clearly highlight patterns because the data is complex and contains multiple attributes, which makes me feel the need for better visualization tools.

2.2 Empathy Map Canvas

User: Healthcare analysts and medical researchers.

Thinks

Which factors increase heart disease risk?

Is smoking strongly related to heart disease?

Which age group is most affected?

Feels

Concerned about increasing heart disease cases.

Needs accurate insights for decision making.

Wants a clear understanding of health data.

Says

Visualization tools help understand data faster.

Lifestyle habits should be analyzed properly.

Does

Analyzes healthcare datasets.

Uses visualization tools like Tableau.

Studies patterns between health indicators and diseases.

Pains

Large datasets are difficult to interpret.

Raw data does not easily show patterns.

Gains

Interactive dashboards for analysis.

Clear visualization of risk factors.

Faster understanding of health trends.

2.3 Brainstorming

During the brainstorming session, the team discussed various ideas for analyzing heart disease data and visualizing the results effectively.

Generated Ideas

Analyze heart disease cases by age category.

Compare smokers vs non-smokers.

Study the relationship between BMI and heart disease.

Analyze the impact of physical activity.

Study the effect of sleep duration.

Compare heart disease cases across genders.

Analyze diabetes and heart disease relationship.

Selected Ideas

The team prioritized ideas based on feasibility and impact.

Selected visualizations for dashboard:

Heart Disease by Age Category

Heart Disease by Gender

Heart Disease by Smoking Habit

Heart Disease by BMI

Heart Disease by Physical Activity

Heart Disease by Sleep Duration

3. REQUIREMENT ANALYSIS:

3.1 Customer Journey Map

User opens the Tableau dashboard.

User selects filters such as age category or gender.

Dashboard displays visualizations related to heart disease.

User analyzes patterns between health indicators and heart disease.

User interprets insights to understand risk factors.

3.2 Solution Requirement

Functional Requirements

Import dataset into Tableau.

Create charts and graphs for analysis.

Implement filters for age, gender, and lifestyle factors.

Display heart disease statistics visually.

Non-Functional Requirements

Easy-to-understand dashboard.

Fast visualization rendering.

Interactive filters for better exploration.

3.3 Data Flow Diagram

Level 0 (Overview)

Dataset → Data Processing → Tableau Visualization → Dashboard Insights

Level 1

CSV Dataset
↓
Data Cleaning and Processing
↓
Data Visualization using Tableau
↓
Dashboard Creation

↓
User Interaction and Analysis

3.4 Technology Stack

Technology, Purpose

Tableau, Data visualization and dashboard creation

CSV Dataset, Data source

Excel, Data inspection and preprocessing

Windows OS, Development environment

4. PROJECT DESIGN:

4.1 Problem Solution Fit

The problem of analyzing complex healthcare data can be solved using visualization techniques. Tableau provides interactive dashboards that help users understand patterns in data quickly. By applying visual analytics to heart disease data, the project helps identify important risk factors.

4.2 Proposed Solution

The proposed solution is an interactive Tableau dashboard that visualizes heart disease data using different charts and filters.

The dashboard will allow users to:

Analyze heart disease distribution across age groups.

Compare heart disease cases between genders.

Study the relationship between smoking and heart disease.

Analyze how BMI and physical activity affect heart disease risk.

4.3 Solution Architecture

Architecture Flow

Dataset (CSV File)
↓
Data Cleaning and Preparation
↓
Data Import into Tableau
↓
Creation of Visualizations
↓
Dashboard Integration
↓
User Interaction and Analysis

5. PROJECT PLANNING & SCHEDULING:

5.1 Project Planning

Phase, Description

Dataset Collection, Obtain heart disease dataset

Data Understanding, Analyze dataset attributes

Visualization Design, Decide charts and dashboard layout

Dashboard Development, Build dashboard in Tableau

Testing, Validate dashboard outputs

Final Report, Prepare project documentation

6. FUNCTIONAL AND PERFORMANCE TESTING:

6.1 Performance Testing

The dashboard was tested to ensure proper functionality.

Testing Criteria

Test Case, Result

Dataset loading, Successful

Chart generation, Successful

Filters working, Successful

Dashboard interaction, Successful

The dashboard performs efficiently and provides quick visual insights.

7. RESULTS:

The project successfully created a Tableau dashboard that analyzes heart disease data and presents insights visually.

Key Observations

Heart disease cases increase with age.

Smoking is strongly associated with heart disease.

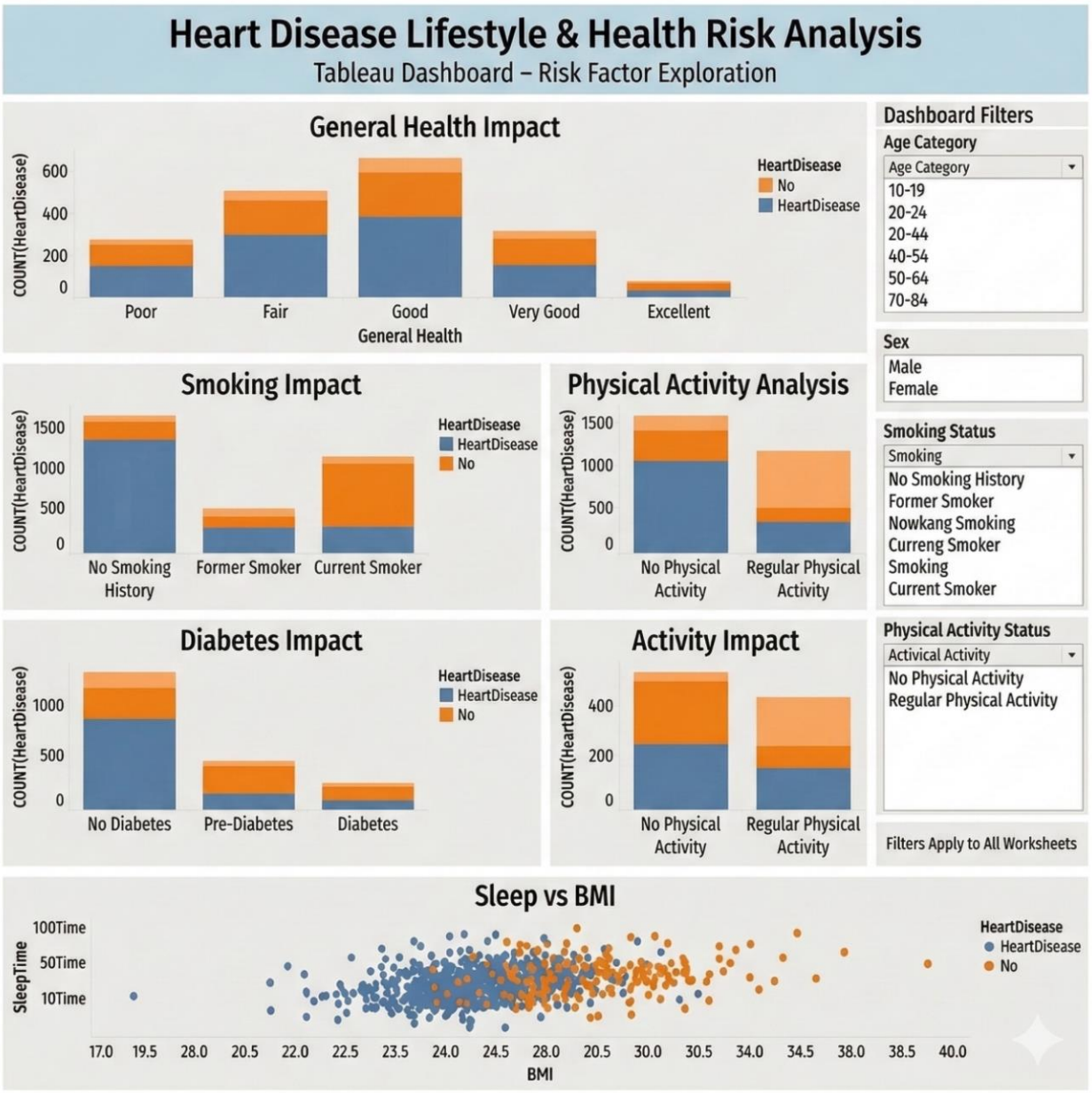
Higher BMI values show higher risk of heart disease.

Physical activity appears to reduce risk factors.

7.1 Output Screenshots

Heart Disease Risk Analysis Dashboard

Dataset Analysis using Tableau



Examples:

Heart Disease by Age Category chart

Smoking vs Heart Disease chart

BMI vs Heart Disease visualization

Final Interactive Dashboard

8. ADVANTAGES & DISADVANTAGES:

Advantages

Easy visualization of complex healthcare data.

Interactive dashboard for better exploration.

Helps identify major risk factors of heart disease.

Supports data-driven healthcare analysis.

Disadvantages

Dataset may not represent all populations.

Limited to available health indicators.

Requires Tableau knowledge for development.

9. CONCLUSION:

The Heart Disease Analysis project successfully demonstrates how data visualization can help understand complex healthcare data. By using Tableau dashboards, patterns and relationships between lifestyle habits and heart disease risk factors can be clearly observed.

The project highlights the importance of factors such as smoking, BMI, age, and physical activity in influencing heart disease risk. Visualization tools make it easier for healthcare professionals to interpret data and make informed decisions.

10. FUTURE SCOPE:

Future improvements for this project include:

Integrating machine learning models for heart disease prediction.

Expanding the dataset to include more health indicators.

Deploying dashboards on web platforms for wider accessibility.

Adding real-time healthcare data analysis.

11. APPENDIX:

Dataset Link

https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view

Source Code

Not applicable (Tableau-based visualization project).

GitHub & Project Demo Link

Github Link: https://github.com/manish43221/Heart_Disease_Analysis
Demo Link: https://drive.google.com/file/d/1DlR_K-Lm6UYpeBjl3SllSEmZgKWBikLV/view?usp=drivesdk